

Problem 8

Under what condition do points $z = x + iy$ lie inside a circle of radius r with center at point $c = a + bi$?

Solution

We need to find the condition for a point z to lie inside (not on or outside) a circle with given center and radius.

Step 1: Geometric Interpretation

A point z lies inside a circle if and only if its distance from the center is less than the radius.

Circle Definition

- Center: $c = a + bi$ (point (a, b) in the plane)
- Radius: $r > 0$
- Point: $z = x + iy$ (point (x, y) in the plane)

Step 2: Distance Formula in Complex Plane

The distance between two complex numbers z and c is given by the modulus of their difference:

$$\text{distance} = |z - c|$$

Step 3: Condition for Interior of Circle

A point z lies **inside** the circle (in the interior, not including the boundary) if:

$$|z - c| < r$$

Step 4: Explicit Form

Substituting $z = x + iy$ and $c = a + bi$:

$$\begin{aligned} |z - c| &= |(x + iy) - (a + bi)| \\ &= |(x - a) + i(y - b)| \\ &= \sqrt{(x - a)^2 + (y - b)^2} \end{aligned}$$

Therefore, the condition becomes:

$$\sqrt{(x - a)^2 + (y - b)^2} < r$$

Step 5: Alternative Forms

Form 1: Using Modulus (Complex Form)

$$\boxed{|z - c| < r}$$

This is the most compact form using complex number notation.

Form 2: Squared Form

Squaring both sides (valid since both sides are non-negative):

$$|z - c|^2 < r^2$$

$$(x - a)^2 + (y - b)^2 < r^2$$

This is often more convenient for calculations.

Form 3: Cartesian Form

$$\sqrt{(x - a)^2 + (y - b)^2} < r$$

Step 6: Related Conditions

For completeness:

Region	Condition
Interior (inside, open disk)	$ z - c < r$
Boundary (circumference)	$ z - c = r$
Exterior (outside)	$ z - c > r$
Closed disk (interior + boundary)	$ z - c \leq r$

Step 7: Example

Example: Find the condition for points inside a circle of radius 3 centered at $c = 2 + i$.

Solution

Here $c = 2 + i$ (so $a = 2, b = 1$) and $r = 3$.

The condition is:

$$|z - (2 + i)| < 3$$

In Cartesian form:

$$(x - 2)^2 + (y - 1)^2 < 9$$

This describes all points (x, y) inside the circle centered at $(2, 1)$ with radius 3.

Step 8: Verification

Let's verify with a specific point. Is $z = 2 + 2i$ inside the circle?

$$|z - c| = |(2 + 2i) - (2 + i)| = |i| = 1$$

Since $1 < 3$, yes, the point $z = 2 + 2i$ is inside the circle.

Is $z = 5 + i$ inside the circle?

$$|z - c| = |(5 + i) - (2 + i)| = |3| = 3$$

Since $3 = 3$ (not < 3), the point $z = 5 + i$ is on the boundary, not inside.

Geometric Interpretation

In the complex plane:

- The set $\{z : |z - c| = r\}$ is a **circle** (1-dimensional curve)
- The set $\{z : |z - c| < r\}$ is an **open disk** (2-dimensional region, excluding boundary)
- The set $\{z : |z - c| \leq r\}$ is a **closed disk** (2-dimensional region, including boundary)

Final Answer

Condition for a point $z = x + iy$ to lie inside a circle of radius r with center $c = a + bi$:

$$|z - c| < r$$

Equivalent forms:

- Complex form: $|z - c| < r$
- Squared form: $|z - c|^2 < r^2$
- Cartesian form: $(x - a)^2 + (y - b)^2 < r^2$
- Expanded: $\sqrt{(x - a)^2 + (y - b)^2} < r$

Key points:

- The inequality is **strict** ($<$, not $\leq$) for points strictly inside
- The modulus $|z - c|$ represents the distance from z to c
- The condition states: distance from center is less than radius